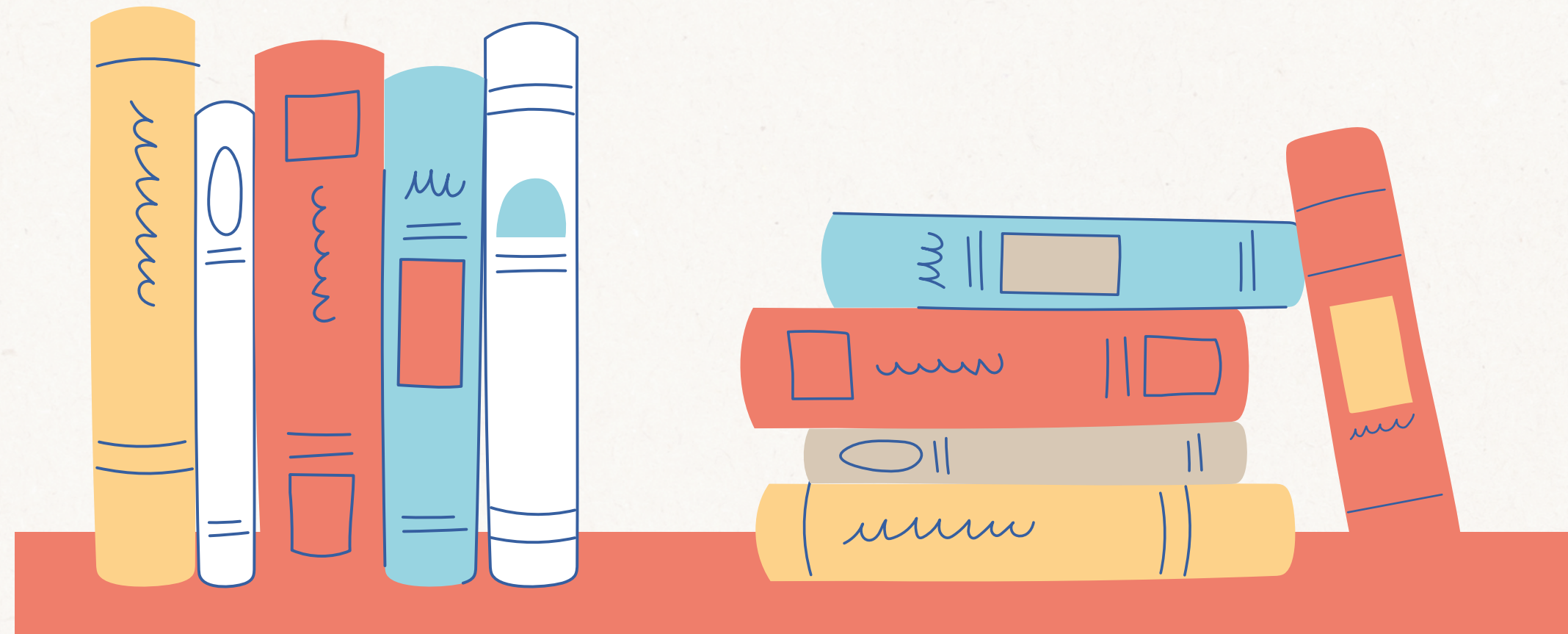


The ReBook

2nd Hand Book
Marketplace

Team Min-Maxers

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Tamjidur Rahman	230042109
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Introducing ReBook



- Buy used books
- Sell used books
- Lend books
- Borrow books
- Review and verification
- Smart Search 
- Wishlist 

User Roles



Lender/Seller

- Upload books
- Set price
- Set lending duration
- Manage listings

Admin

- Verify users
- Moderate listings
- Handle disputes

Buyer

- Search books
- Purchase books
- Review sellers
- Wishlist

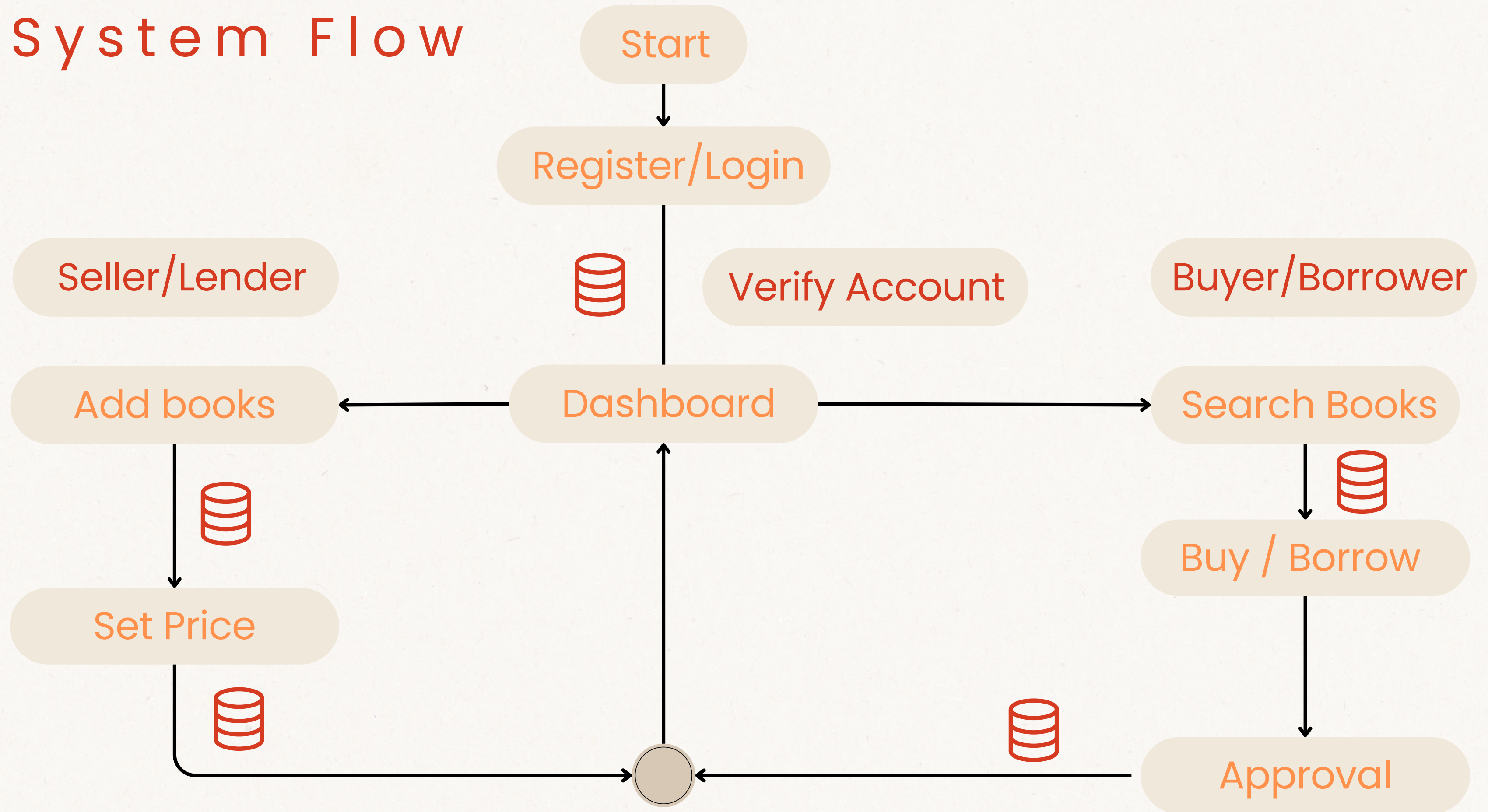
Database Design Planning

Entity	Target Schema Mechanics & Constraints
User	Role-Enforced Identity: Manages 3-level access control (Buyer/Seller/Admin)
Book	Uses a Natural Primary Key (ISBN-13) to globally unique-index editions and optimize structural joins.
Genre/Book Condition	Lookup Tables to Enforce Strict Third Normal Form and remove transitive dependencies
Listing	Bind sellers to books, using conditional CHECK constraints to ensure pricing data matches the transaction type
SaleTransaction	Atomic Ledger: Managed by BEFORE UPDATE triggers to instantly lock rows and prevent concurrent purchasing race conditions

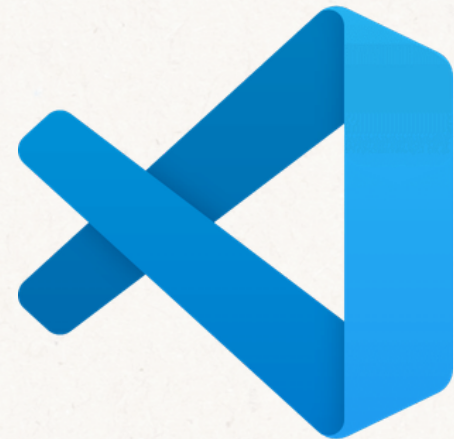
Database Design Planning

Entity	Target Schema Mechanics & Constraints
LendingRecord	A time-tracking table that uses triggers to figure out late fees based on how much time has passed.
Review	Conditional Constraint Router: Utilizes a CHECK constraint preventing orphaned or duplicate feedback mappings.
Audit_Log	Security & Verification Ledger: An immutable history table populated strictly via a row-level AFTER UPDATE trigger to capture financial and wallet changes.
Active_Listings_Cache	Performance Optimization View: A database view utilizing complex LEFT JOIN operations across the User, Book, and Listing tables to serve instant catalog searches.

System Flow



Tools and Technology



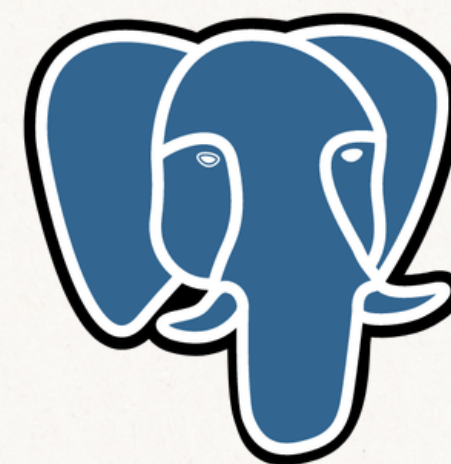
IDE



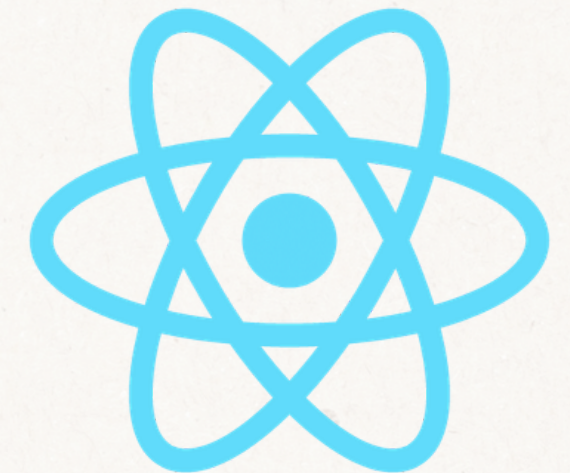
DOCUMENTATION



BACKEND



DATABASE



FRONTEND



VERSION CONTROL

Thank you

Any Questions?

